

DEEPAK G

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Education

Indian Institute of Technology, Kharagpur

B.TECH.(Hons.) in Aerospace Engineering

Aug. 2023 – May 2027

8.8/10

Relevant Coursework

- Programming and Data Structures
- Linear Algebra
- Advanced & Transform Calculus
- AI Foundations and Applications
- AI/ML for Robot Autonomy
- MPC with Applications
- Automatic Control of Aircraft
- Embedded Control and Software
- Aircraft Stability and Control
- Low & High Speed Aerodynamics
- Aerospace Propulsion Systems
- *Advanced Gas Dynamics

mentioned with * are in progress

Publications

- **Deepak G**, Rakesh Sahoo, Manoranjan Sinha, “Autonomous Spacecraft Rendezvous and Docking via Hierarchical Deep-Reinforcement Learning and Model Predictive Controller,” *AIAA SciTech Forum 2027*, Accepted for presentation.

Research Experience

Caltech SURF 2026 | AMBER Lab

May 2026 – Present

Research Fellow — Advised by Prof. Aaron D. Ames, Dr. Yana Lishkova, Dr. Joe Moeller & Dr. Pedro Roque

- Developed a tube-based convex MPC approach for optimal control of the full nonlinear 6-DOF spacecraft dynamics.
- Derived a Difference-of-Convex (DC) representation of the dynamics analytically, using ICNNs & convex polynomials.
- Integrated a Control Barrier Function (CBF)-based safety filter to develop a multilayered safe control architecture.
- Validated the approach in Gazebo simulations and on the planar robotic testbed ATMOS.

CSIR-National Aerospace Laboratories (NAL)

May 2025 – Jul. 2025

Research Intern - Advised by Dr. Guruganesh R

- Conducted a literature survey on global high-altitude airship projects to identify suitable modeling assumptions.
- Derived 6-DOF equations of motion to account for added mass and inertia tensors unique to buoyant vehicles.
- Developed the complete nonlinear flight dynamics model of a high-altitude airship in MATLAB/Simulink.
- Identified trim conditions under suitable assumptions and evaluated open-loop stability of the linearized model.

Projects

Safe Reinforcement Learning for Fixed-Wing UAV Control

Dec. 2025 - Feb. 2026

Advised by Prof. Harikumar Kandath, Robotic Research Centre, IITH

- Developed and validated a nonlinear 6-DOF fixed-wing UAV model in Python for reinforcement learning-based control.
- Designed a Gymnasium environment and trained a PPO agent for altitude tracking through reward shaping.
- Integrated a Control Barrier Function-based safety filter to enforce stall avoidance constraints during RL-based control.

3D Encirclement and Capture with unknown evader strategies | Research project

Nov. 2025 - Present

Advised by Prof. Ashish Hota, Dept. of Electrical Engineering IIT Kharagpur

- Investigated a volume-based approach for encirclement of an evader in a three-dimensional unbounded environment.
- Designed a hybrid pursuit strategy based on geometry for finite-time capture and encirclement of the evader.
- Developing Control Barrier Function Formulations for Guaranteed Encirclement.

Schlieren Imaging & Supersonic Aerodynamic Analysis | Research Project

Sep. 2025 - Dec. 2025

- Conducted high-speed flow visualization of Concorde and Sukhoi Su-57 nose sections at Mach 2.2 across varying angles of attack (0° to 10°). Prepared scaled CAD models of the nose sections and fabricated test models via CNC machining.
- Compared shock angles from Schlieren images against theoretical predictions from the Taylor-Maccoll theory.
- Analyzed the impact of the chined geometry of Su-57 on shock curvature, identifying deviations from the Concorde.

PID Controller for Quadcopter Altitude Control | Self-Project

Nov. 2024

- Modeled vertical dynamics and designed an altitude control system in Simulink to achieve stable altitude hold.
- Simulated real-world conditions by incorporating actuator saturation and sensor noise to assess controller robustness.
- Iteratively tuned PID gains to minimize overshoot, improve settling time, and reduce steady-state tracking error.

Technical Skills

Languages: Python, C, MATLAB, LaTeX

Tools and Libraries: Numpy, Pandas, Matplotlib, PyTorch, Gymnasium, Stable-Baselines3, YALMIP, CVXPY, Git

Software: PyBullet, ROS 2, Optitrack Motive, Simulink, SolidWorks, Fusion 360, ANSYS

Leadership / Extracurricular

- Member of the Mechanical Subsystem at TeamTrak. Contributed to the design of a miniature locomotive to participate in the IMechE Railway Challenge (UK)
- Iteratively designed and tested three RC aircraft to meet Boeing’s mission-specific payload requirements. Ranked in the Top 30 nationally in the Boeing National Aeromodelling Competition.
- Organised the 7th edition of Local Startups’ Meet Chennai 2024, facilitating participation of 30+ startups, 20+ investors, and 5+ VCs.